

SIMON MARIANI

Software Engineer

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WORK EXPERIENCE

Software Engineer

Univia

- Jul 2023 - present (Full-time)
- Eindhoven, NL/Berlin, DE
- Designed **microservice-based architectures** and built/deployed large-scale backend services using **Docker**, **Docker Compose**, **Kubernetes**, and **Helm**, implementing **CI/CD pipelines** in **GitHub Actions** to ensure **reliable, production-ready systems**, reduce deployment time, and improve **release reliability**.
 - Served as cross-project technical point of contact, **coordinating API access** and **scheduled data ingestion** using various tools, including **N8n**, **Apache Airflow** and **shell scripting + cron jobs**.
 - Deployed and operated a **multi-node Kubernetes cluster** on **bare-metal infrastructure** using **Proxmox VE** for virtualization, improving **scalability and stability**.
 - Developed **AI-powered tools** for clients and internal products, improving efficiency and functionality using **Python** libraries including **NumPy**, **Pandas**, **Scikit-learn** and **PyTorch**.
 - Developed a user-friendly platform with **ReactJS** and **Django**, providing streamlined deployment and execution of modular applications on a **Kubernetes cluster**.
 - Developed a real-time platform for municipalities (**Meetnet**) with **ReactJS**, **ExpressJS**, and **time-series databases**, implementing **multi-region data ingestion pipelines** and managing **300+ GB** of **sensor data** to enable **reliable, actionable insights**.
 - Developed a **retrieval-augmented** chatbot with file system integration and **vector-based search** using **Django**, **React** and **LLM clients** in **Python**, helping users to quickly search through large amounts of information.
 - Helped architect a **scalable platform** for hardware module management and implemented a frontend in **VueJS** for interactive platform management.
 - Created and managed an internal **AI knowledge bank** to reduce project startup time and improve company-wide knowledge retention, using **PyTorch**, **Keras**, **Scikit-learn**, **Statsmodels** and **Pandas**.
 - Built a **Computer Vision** tool for automatic reading and processing of complex manufacturing components using **OCR** and **classic Computer Vision techniques**.
 - Analyzed **timeseries data** with **Python** and **machine learning models** to detect anomalies and uncover meaningful patterns for user decision-making.

Computer Vision Research Intern

ViNotion

- Nov 2021 - Jul 2022 (Full-time)
- Eindhoven, NL
- Researched **Segmentation Networks** and **Out-of-Distribution (OOD) Detection**, optimizing **Deep Learning** models on **large image datasets** and improving OOD benchmark performance; codebase available on **GitHub** using **Python** and **PyTorch**.
 - The findings of the research were **published at a recognized academic conference**, furthering contributions to the field of computer vision and can be found here: **The Effect of Covariate Shift and Network Training on Out-of-Distribution Detection**.
 - The results of the study were implemented for **Image Segmentation on proprietary datasets**, which improved their AI capabilities.

Programming Camp Coordinator

Vinea

- June 2019 - Aug 2019 (Part-time)
- NL
- Delivered **masterclasses** on **game design** principles and **web development**, introducing young learners to fundamental concepts in software development.

ABOUT ME

Software engineer specializing in backend, full-stack, and AI development, building scalable, production-ready applications with cloud infrastructure and machine learning pipelines.

LANGUAGES

- English - Fluent
- Dutch - Fluent
- German - Intermediate
- Spanish - Intermediate
- Chinese - Elementary

TECHNICAL SKILLS

Programming Languages

- Python
 - JavaScript
 - TypeScript
 - HTML/CSS
- C#
 - C++
 - C
 - Clingo/Prolog

Backend Frameworks

- Django
 - ExpressJS
- Flask
 - FastAPI

Frontend Frameworks

- ReactJS
 - VueJS
- Figma

Deployment Tools

- Docker
 - Docker Compose
 - Kubernetes
 - Helm
 - Kustomize
 - Prometheus
 - Grafana
- Loki
 - Promtail
 - Git
 - GitHub Actions
 - N8n
 - Apache Airflow

Databases

- PostgreSQL
 - MySQL
 - MariaDB
 - TimescaleDB
 - InfluxDB
- Milvus
 - FAISS
 - MongoDB
 - Redis

Cloud Platforms

- Azure
 - OVH
 - Digital Ocean
- AWS
 - Scaleway

Master of Science in Artificial Intelligence

📅 Sep 2020 - July 2022

📍 Amsterdam, NL

- Earned the distinction **cum laude** for academic excellence.
- Solved complex coding problems using **Python**, optimizing algorithms for efficiency.
- Completed a rigorous, technical program that included **Deep Learning, Machine Learning, NLP, Reinforcement Learning, CV, and Active Learning**.
- Collaborated with a team of students to conduct a **reproducibility study** using **Python** and **PyTorch**, validating its findings and **extending the original work** to assess the degree of reproducibility. The results can be found here: [*Generative Causal Explanations of Black-box Classifiers*](#).

Tsinghua University

📅 Sep 2019 - Feb 2020

📍 Beijing, CN

- Developed a **Distributed Neural Network training system** across three **Linux servers**, working with **multi-node orchestration** and speeding up model training.
- Completed courses on **Human-Computer Interaction** and **Distributed Database Systems**, gaining practical experience in **user-centered design** and **scalable data architectures**.

Utrecht University

📅 Sep 2017 - July 2020

📍 Utrecht, NL

- Developed strong programming skills in **Python**, **C#**, and **Prolog** through coursework focused on **software testing**, **verification**, and **algorithm design**.
- Applied multiple **Supervised Machine Learning** methods in a Bachelor's thesis to analyze and profile serial killers, demonstrating **data-driven modeling** and **evaluation skills**. The results can be found here: [*Profiling Serial Killers Using Multiple Supervised Machine Learning Approaches*](#).

Personal Website

- Developed a personal website with an integrated chatbot and retrieval-augmented generation system as well as informative graphs using **ReactJS**, **FastAPI** and **Vector Databases** available at simonmariani.com.

AI Tools

- Numpy
- pandas
- matplotlib
- Scikit-learn
- PyTorch
- CUDA
- LLM APIs
- Huggingface
- Langchain
- Keras
- Tensorflow

- MLOps
- Deep Learning
- Machine Learning
- Computer Vision
- NLP
- Reinforcement Learning
- Anomaly Detection
- Recommender systems
- Time Series Analysis
- OOD-detection
- Active Learning

Models & Architectures

- CNNs
- ML Models
- RNNs/LSTMs
- Transformers
- VAEs
- GANs
- Bayesian Models